

EduGenie – Google Gemini Powered Learning Assistant

Scalability & Future Plan

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Current System Limitations

Limitation	Impact	Priority to Address
No persistent accounts/progress	Learning history is not retained.	High
Single-language current implementation	Limits accessibility for some learners.	Medium
LLM output may occasionally be inaccurate	Requires user verification for important academic content.	High

Scalability Plan

Aspect	Current State	Proposed Upgrade / Solution
User Load	Lightweight single-service application	Containerized deployment, horizontal scaling and rate limiting.
Data Storage	Stateless; no database	Add PostgreSQL or similar persistent store for accounts/progress.
Performance	Network/model latency dependent	Caching, async optimization, monitoring and load testing.
Security	Environment-based API key	Authentication, authorization, secret management and audit logging.

Future Roadmap

Phase	Planned Enhancement	Expected Impact
Phase 2	User accounts and progress tracking	Personalized continuity across sessions.
Phase 3	Multi-language and spaced repetition	Broader accessibility and stronger revision support.
Phase 4	LMS integration, mobile and voice interfaces	Wider deployment and richer learning experience.